**WHITE PAPER**

Version 8

# The Underground Waste Refinery

*A Modular, Serviceable, Distributed System for Thermal Shock Separation*
*and Pyrolysis Conversion of Municipal Solid Waste*

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# Executive Summary

The Underground Waste Refinery (UWR) is a decentralized waste processing system designed to replace conventional underground waste containers. It integrates proven industrial technologies—steam explosion, surfactant cleaning, drying, near-infrared sorting, electrostatic separation, and catalytic pyrolysis—into a factory-built, swappable module that can be lowered into a prepared underground vault and automatically connected to utilities. Routine maintenance is performed through surface-accessible hatches; the entire module can be lifted out and replaced with a refurbished spare for major overhauls, minimizing site downtime.

The system processes mixed municipal solid waste (MSW) at the neighborhood scale, producing:

- Liquid fuel from recovered plastics,
- Organic broth transported to a centralized anaerobic digestion facility (on-site AD available as a site-specific add-on),
- Clean recyclates (metals, glass),
- Syngas used to sustain the process.

Layered safety measures, comprehensive noise and vibration control, and a modular regulatory pathway make the UWR suitable for unattended urban deployment. This document presents the complete technical specification, energy balance, economic model, and implementation roadmap for Version 8—the final concept design ready for prototyping.

# 1. Introduction

## 1.1 The Problem

Municipal solid waste, especially mixed household garbage, imposes high logistical and environmental costs. Collection and transportation account for a large share of municipal budgets and carbon emissions. Centralized recycling and waste-to-energy facilities require long haul distances and often reject contaminated streams. A distributed solution that processes waste at the point of generation can dramatically reduce hauling costs, recover valuable materials, and lower the carbon footprint.

## 1.2 The Concept

The UWR is a self-contained unit installed in an underground vault at existing or new waste container sites. It performs a multi-stage treatment:

1. Shredding & accumulation
2. High-pressure steam explosion (thermal shock) to hydrolyze organics, wood, and textiles
3. Gravity discharge into a wash-filter box for broth separation and solids washing
4. Hot water and steam jet washing with surfactant to remove fats and residual organics
5. Vertical heated screw drying under nitrogen, using waste heat
6. Near-infrared (NIR) sorting to reject PVC, with integrated metal removal
7. V-convergence conveyor lip separation to isolate glass and inerts from plastics

8. Fine grinding to uniform plastic flakes
9. Airlock system with nitrogen purge to safely feed clean plastics into pyrolysis
10. Catalytic pyrolysis of the plastic fraction into liquid fuel and syngas
11. Energy recovery — syngas used to sustain the process, waste heat recycled for drying and water heating

All process equipment is housed in a factory-built steel module that can be lowered into a prepared vault and automatically connected. Routine service is done through surface-level hatches; the entire module can be swapped for major overhauls.

# 2. System Architecture

## 2.1 Overall Layout

### Vault Dimensions — Retrofit vs. New Deployment

Vault dimensions are not fixed. Two deployment scenarios are anticipated:

- Retrofit deployment: The module is installed at an existing underground container site. Vault dimensions are constrained by the existing excavation, foundation, and utility connections. A site survey is required before module specification.
- New deployment: The vault is purpose-built to reference dimensions optimized for the module — internal dimensions 6 m × 2.5 m × 5.5 m, with a steel guide frame, utility docking plate, and an integrated service corridor (0.6 m) on one side.

In both cases the module contains all process equipment in a structural steel frame with a quick-connect docking plate at the bottom, surface-accessible hatches, and crane lifting points. Final module dimensions are confirmed during the 3D CAD phase and will reflect the deployment context.

Above-ground support house (optional): Small enclosure (2 m × 2 m) housing the steam boiler, nitrogen generator, and electrical panel if not integrated into the module.

## 2.2 Subsystem 1: Collection & Primary Shredding

- Lid: Sealed, lockable. A weatherproof camera above the inlet records each deposit for accountability and hazardous item monitoring; footage is retained and remotely accessible by the operator.
- Inlet restrictions: Opening sized to prevent insertion of large hazardous items.
- Slow-speed shredder: Mounted at the top of the module, directly below the lid. Activates briefly after lid closure; motor current feedback prevents jams. Output size 50–100 mm.
- Metal detector & reject flap: Detects batteries and small gas cartridges; diverts to a lockbox. CO sensor in the shredder compartment provides early warning of thermal runaway; anomalous current spike detection also flags potential battery events.
- Separate hazardous waste bin: Locked compartment at the site for batteries, gas cartridges, and chemicals.

## 2.3 Subsystem 2: Thermal Shock (Steam Explosion) Vessel

The pressure vessel is an inclined cylindrical vessel rated for 25 bar (2.5 MPa) and 230 °C. Its inclined orientation serves the entire loading and discharge sequence passively — waste slides in by gravity, and slurry drains out by gravity at the lower discharge end.

- Loading: An inclined chute connects the shredder output to the vessel inlet. The chute's lower end is hinged and normally hangs loose. As the pneumatic inlet door closes upward, it lifts and supports the hinged chute end from below, folding it flush and clear of the door seal in a single motion — no separate chute actuator is required.
- Inlet door: Closes upward pneumatically, sealing the vessel for pressurization. Its closing motion simultaneously folds the loading chute as described above.
- Steam injection: From the boiler (support house or integrated). Rapid pressurization to operating conditions.
- Soak: 30–120 seconds at pressure, duration optimized via prototyping.
- Treatment verification: An ultrasonic level sensor mounted on the vessel wall measures fill level before steam injection and again after decompression and settling. The ratio of volume reduction is used as a proxy for treatment effectiveness — well-processed material collapses significantly as cellular structures rupture. If the reduction ratio falls below the control threshold, one or more additional explosion cycles are triggered automatically before discharge. The threshold is a calibration output from prototype testing and can be adjusted adaptively by the AI monitoring system based on observed downstream solids quality.
- Staged decompression: A sequenced valve manifold releases pressure in 3–5 controlled steps over 10–30 seconds, reducing acoustic impulse. An anti-surge expansion vessel acts as an acoustic buffer.
- Discharge door: Closes downward pneumatically on the lower end of the inclined vessel. When opened, the inclined geometry directs the slurry toward the discharge opening by gravity. A water or steam flush assist ensures complete vessel evacuation while simultaneously rinsing the vessel interior.

## 2.4 Subsystem 3: Heat Recovery, Vacuum & Condensate Management

- Single expansion chamber: A large (2–3 $m^3$) vessel captures the bulk of flash steam after decompression. Condensation recovers heat and generates a partial vacuum for downstream filtration. The large volume also provides acoustic smoothing, supplementing the anti-surge vessel upstream. The expansion chamber is connected to the wash-filter box via a pressure relief path, ensuring that any overpressure event resulting from a valve failure is safely vented to the expansion chamber rather than into the open wash-filter box enclosure.
- Condensate trap (knockout pot): A compact vessel — sized as small as the available module space permits — is installed downstream of the expansion chamber. Its sole function is to capture residual liquid carryover before it reaches the vacuum pump. A high-level float switch triggers an alarm before overflow can occur. Condensate drains to the water treatment buffer tank.
- Supplemental vacuum pump: A small mechanical vacuum pump (liquid-ring or rotary vane) provides reliable, regulated vacuum for filtration independent of steam condensation rate.

## 2.5 Subsystem 4: Wash-Filter Box & Primary Cleaning

After decompression, the slurry is discharged from the explosion vessel into a dedicated wash-filter box positioned almost at ground level, directly below the vessel discharge door. This separation of functions is deliberate: the explosion vessel is thick-walled and rated for 25 bar with minimal penetrations, while the wash-filter box operates at atmospheric pressure, making it straightforward to fabricate, access, and clean.

The wash-filter box is an open-topped fabricated steel box with sharply angled internal walls that channel the incoming slurry and solids downward onto the filtration mesh. It is not a pressure vessel — it is a robust, open industrial enclosure designed for easy access and cleaning. A pressure relief path connects it to the expansion chamber to safely manage any residual pressure from the discharge event.

- High-pressure jets: Overhead water and steam jets wash the solids as they arrive, dislodging residual organics and fats. Alkyl polyglucoside surfactant is dosed in the first wash zone; a clean hot water rinse follows in the second zone.
- Filtration mesh: The mesh floor or inclined mesh screen (0.5–2 mm) allows vacuum-assisted drainage of the organic broth downward into the collection system. The vacuum drawn by the expansion chamber and supplemental pump pulls broth through the mesh efficiently.
- Screw conveyor discharge: A screw conveyor at the base of the wash-filter box actively conveys the filtered and washed solids out toward the vertical dryer. The angled walls funnel all solids toward the screw intake, preventing accumulation.
- Camera / AI monitoring: A camera positioned above the screw discharge observes solids quality as they are conveyed out. If excessive unfiltered organic matter is detected, the system triggers additional washing cycles for the current batch. It also adjusts the volumetric reduction threshold upward for the next explosion cycle, so the system self-calibrates over time to variations in waste composition without operator intervention.
- Self-cleaning mesh: Rotating brush and backflush mechanism prevents mesh blinding during operation.
- Organic broth: Collected broth is the primary output of this stage. The baseline assumption is transport to a centralized anaerobic digestion facility. An on-site AD unit is available as a site-specific add-on where biogas generation is desired locally.
- Wash water is collected and sent to the water treatment loop.

## 2.6 Subsystem 5: Drying

### Vertical Heated Screw Dryer

- Washed solids enter from the bottom of a vertical screw conveyor.
- Heated jacket: Waste heat from boiler flue gas and pyrolysis cooling raises temperature to 80–100 °C. Electric backup heater included for startup and cold conditions.
- Nitrogen purge: Continuous low-flow nitrogen maintains an inert atmosphere and carries away moisture.
- Self-regulating: Wet material stays low; dry material exits at the top lateral outlet.
- Target moisture at exit: <5% for reliable NIR performance downstream.

## 2.7 Subsystem 6: Sorting & Separation Train

The sorting and separation train is a single continuous conveyor system that takes dry mixed solids from the dryer exit and delivers clean, ground plastic to the weighing funnel. It performs NIR sorting, metal removal, glass/inert separation, and fine grinding in a logical sequence, with each stage set up by the geometry of the previous one.

### 6.1 Concave Rubber Conveyor — Spreading & NIR Zone

Dry solids from the dryer exit drop onto a wide concave rubber conveyor belt. A bump actuator underneath the belt fires a single sharp impulse just after the material drops, spreading fragments across the concave profile into a single layer. The concave belt cross-section prevents fragments from sliding off the edges. This spread single-layer presentation is required for reliable NIR detection.

- Overband magnet: Positioned above the spread belt, removes ferrous metal fragments from the single layer cleanly and effectively.
- Electrostatic deflector: Positioned along the conveyor, deflects non-ferrous metal fragments (aluminum and other conductors) sideways off the belt into a collection bin. Operating on a spread single layer gives better separation efficiency than operating on a falling curtain.
- NIR PVC sorter: Near-infrared spectrometer scans the spread material over the 1.5 m active belt zone. Air jets eject identified PVC flakes into a separate bin. Black plastics (carbon-black filled, invisible to NIR) are diverted to a separate disposal stream, with a LIBS upgrade path available. With ferrous and non-ferrous metals already removed, the NIR scan operates on a clean mixed plastic and glass/inert stream.

### 6.2 V-Convergence Zone — Stream Concentration

After the NIR zone, the conveyor geometry transitions. The belt rollers progressively reduce in width and side actuators gently nudge material inward from both edges, converting the wide spread mat into a narrow concentrated stream along the belt centerline. By the time the material reaches the conveyor end, it is a single narrow curtain of mixed fragments — glass, inerts, and plastic flakes — moving horizontally at conveyor speed.

### 6.3 Lip Gap Separator — Glass/Inert Removal

At the end of the conveyor, a short gap is introduced before the knife mill inlet. A horizontal slotted vacuum pipe runs across the full conveyor width just above this gap, generating a gentle, evenly distributed low-pressure zone. The suction is calibrated to provide just enough upward bias to deflect light plastic flakes onto the knife mill inlet above, while glass fragments and heavy inerts carry sufficient momentum and mass to arc over the gap and fall into the collection bin below. The vacuum level required is minimal — the V-convergence has already concentrated the stream and the conveyor's own momentum does most of the work. The knife mill's blade rotation may generate sufficient self-aspiration to assist or replace the dedicated vacuum source; this will be evaluated during prototype testing.

### 6.4 Knife Mill

Receives the deflected plastic stream directly from the lip gap. Grinds clean plastic — with metals and glass already removed — to uniform 3–5 mm flakes. The narrow concentrated feed from the V-convergence provides a relatively steady input, smoothing mill load.

### 6.5 Covered Screw Conveyor to Weighing Funnel

A sealed screw conveyor transports ground plastic flakes from the knife mill outlet to the weighing funnel, covered to prevent contamination and contain any fine dust generated by grinding.

### 6.6 Secondary Electrostatic Separator — Final Purity Gate

A final electrostatic separator is positioned at the weighing funnel inlet, acting on the plastic stream as it drops from the screw conveyor into the funnel. Any residual fine non-ferrous metal fragments are deflected away before the plastic enters the nitrogen-purged airlock system. This is the final purity gate before pyrolysis — positioned at the most critical point in the process.

## 2.8 Subsystem 7: Pyrolysis Feed System (Smart Airlock) & Reactor

- Weighing funnel: Suspended from ceiling cables with load cells; flexible bellows isolates it from upstream vibration. Accumulates ground plastic until batch mass threshold is reached.
- Top valve: Pneumatic gate valve closes during the airlock cycle.
- Nitrogen purge: PSA unit supplies nitrogen; a buffer tank smooths peak demand across dryer, airlock, and reactor.
- Bottom valve: Opens to drop the batch into the reactor.
- Control logic: Batch mass threshold combined with energy availability triggers purge and feed.
- Pyrolysis reactor: Heated vessel (400–500 °C), inert atmosphere, zeolite catalyst protected by a NaOH guard bed upstream to neutralize residual HCl. Quasi-continuous operation via small buffer hopper.
- Condenser: Liquid fuel collected; non-condensable syngas routed for process heating.
- Syngas utilization: Powers reactor, boiler, and dryer; excess available for electricity generation.
- Char management: Char bin accessible via surface hatch; contents characterized for heavy metals before disposal or reuse. Hazardous char disposed of via licensed contractor.

## 2.9 Subsystem 8: Water Treatment Loop

- Grease trap: Captures bulk fats and oils from wash-filter box drainage.
- Buffer tank: Dampens peak flow from washing jets and condensate drainage.
- Dissolved air flotation (DAF): Removes emulsified fats and fine particles.
- Membrane bioreactor (MBR): Capacity 1,500–2,500 L/day, sized for the reference throughput of 1–2 batches/day. Biological treatment reduces COD and BOD and breaks down residual surfactant.

- Foam control: Mechanical defoamer or antifoam dosing.
- Membrane cleaning: Periodic chemical cleaning (every 1–4 weeks) accessible via service hatch.
- Treated water: Disinfected (UV) and reheated using waste heat for recirculation to the wash-filter box jets. Make-up water added as needed; sewer discharge limited to emergency overflow.
- Biofilm maintenance: During low-frequency operation, a small continuous substrate feed maintains microbial viability.

### 2.10 Subsystem 9: Controls & Monitoring

- PLC/SCADA system: Controls all sequences, monitors sensors, logs operational data.
- Remote monitoring: Alerts operators to maintenance needs or alarms.
- Control network security: Isolated from the public internet; access restricted to authorized personnel.

### 2.11 Subsystem 10: Noise & Vibration Control

- Staged decompression valves and anti-surge vessel.
- Anti-vibration mounts (rubber-steel sandwich) for shredder, knife mill, compressor, fans, and pumps.
- Knife mill acoustic enclosure (double-wall with absorption lining).
- Variable speed drives for compressor and fans.
- Reactive silencers on all vents.
- Constrained layer damping on interior vault walls.
- Scheduling logic: Loud operations (shredder, knife mill, compressor at full speed, steam explosion) restricted to daytime hours (7 am – 10 pm).

## 3. Process Flow

12. User deposits waste → camera records deposit → lid opens → shredder (daytime only) → metal detector rejects hazards to lockbox.
13. Shredded waste slides down inclined chute into pressure vessel → inlet door closes upward, folding chute end → staged steam explosion → ultrasonic volume reduction check → additional cycles if below threshold → discharge door opens downward → slurry discharged by gravity and steam/water flush into wash-filter box below.
14. Wash-filter box: high-pressure jets wash solids → vacuum-assisted broth drains through mesh → screw conveyor moves solids toward dryer → camera/AI monitors solids quality, adjusts washing and next-batch explosion threshold adaptively → broth transported to centralized AD facility (on-site AD optional).
15. Solids → vertical heated screw dryer (80–100 °C, $N_2$) → exit at <5% moisture.
16. Concave rubber conveyor: bump actuator spreads fragments → overband magnet removes ferrous metals → electrostatic deflector removes non-ferrous metals → NIR rejects PVC.
17. V-convergence zone concentrates stream → lip gap separator: glass and inerts drop to bin, plastics deflected onto knife mill.

18. Knife mill grinds to 3–5 mm → covered screw to weighing funnel → secondary electrostatic final purity gate at funnel drop.
19. Weighing funnel accumulates → airlock ($N_2$ purge) → pyrolysis reactor.
20. Liquid fuel stored; syngas powers process; char collected for characterization and disposal.

# 4. Technical Specifications

## 4.1 Throughput Basis

The reference case assumes a service population of 200–250 households (~500–625 persons at 2.5 persons/household), depositing waste at 0.5 kg/person/day, giving approximately 250–310 kg/day total waste input. At a typical MSW plastic content of 15–20% by mass, this yields 37–62 kg of plastic per day — consistent with the 50 kg/batch, 1–2 batches/day assumption used in the energy and economic models.

For smaller deployments (e.g., 50-household retrofit), plastic throughput drops to approximately 9–12 kg/day. The pyrolysis reactor operates in low-frequency batch-accumulation mode and fuel revenue is reduced proportionally. The economic model in Section 6 applies to the reference 200–250 household case; smaller deployments require site-specific recalculation.

| Component | Parameter | Value / Notes |
|---|---|---|
| Module dimensions | L×W×H | Confirmed at 3D CAD stage; reference ~5.5 m × 2.2 m × 5.0 m (new deployment) |
| Vault internal | L×W×H | Reference 6 m × 2.5 m × 5.5 m (new deployment); retrofit sites surveyed individually |
| Shredder | Power | 1–2 kW, output 50–100 mm |
| Steam vessel | Rating | 2–3 m³, 25 bar, 230 °C, inclined, two-door |
| Wash-filter box | Type | Atmospheric, open-top fabricated steel, angled walls, mesh floor, overhead jets, screw discharge |
| Sorting conveyor | Configuration | Concave rubber belt, bump actuator, overband magnet, electrostatic deflector, NIR zone, V-convergence, lip gap separator |
| NIR sorter | Belt length | 1.5 m active zone, air jet rejection |
| Lip gap separator | Mechanism | Slotted vacuum pipe, low pressure differential; glass/inerts drop, plastics deflected to knife mill |
| Knife mill | Output | 3–5 mm flakes; fed after metal and glass removal |

| | | |
|---|---|---|
| Electrostatic separators | Voltage | 20–40 kV; conveyor-mounted (non-ferrous) + funnel drop (final purity gate) |
| Vertical dryer | Height | 3–4 m, heated jacket, $N_2$ purge |
| Water treatment | MBR capacity | 1,500–2,500 L/day including DAF |
| Nitrogen generator | PSA | 50–100 L/min, ≥99% purity, buffer tank |
| Pyrolysis reactor | Batch size | 20–50 kg, quasi-continuous |
| Liquid fuel yield | — | 50–70 wt% of clean plastic input |
| Syngas yield | — | 10–30 wt% of clean plastic input |
| Reference service population | — | 200–250 households |

## 5. Energy Balance

Estimates based on the reference throughput: 200–250 households, 2 batches/day, ~50 kg plastic per batch. Waste heat recovery estimated at 35% for boiler flue gas and 40% for pyrolysis cooling. The replacement of a full pneumatic separation column with the low-energy lip gap separator reduces fan power consumption materially relative to earlier versions.

| Item | Energy per Batch | Notes |
|---|---|---|
| Boiler (thermal, gross) | 80 kWh | Steam generation |
| — Recovered waste heat | −28 kWh | Flue gas & expansion chamber |
| — Net boiler demand | 52 kWh | |
| Drying (latent) | 45 kWh | Evaporating ~60 kg water |
| — Recovered waste heat | −30 kWh | Boiler flue & pyrolysis cooling |
| — Electric backup | 15 kWh | If waste heat insufficient |
| Wash water reheating | 23 kWh | ~1,000 L from 30 °C to 70 °C |
| — Recovered waste heat | −23 kWh | Pyrolysis cooling & dryer exhaust |
| — Net | 0 kWh | Balanced by waste heat |
| Motor electricity | 16 kWh | Shredder, conveyors, compressor, fans, pumps, controls |
| **Total external energy** | 83 kWh | Before syngas credit |
| Syngas from plastic | −40 kWh | 15 kg @ 17 MJ/kg = 71 kWh; 56% used for process heating |
| **Net external energy** | 43 kWh | |

The system is not fully self-sufficient but consumes significantly less external energy than conventional collection and processing. The value of produced fuel and recyclates, combined with avoided hauling costs, makes the system economically viable at the reference scale.

# 6. Economic Model

## 6.1 Capital Costs (Scaled Production)

| Component | Cost (€) |
|---|---|
| Module fabrication, steel frame, hatches | 30,000 |
| Steam vessel (25 bar, PED certified, two-door) | 25,000 |
| Boiler (compact, integrated) | 15,000 |
| Wash-filter box, cleaning jets, mesh, screw | 12,000 |
| Sorting conveyor system (full train) | 45,000 |
| Vertical dryer and conveyors | 18,000 |
| Knife mill | 8,000 |
| MBR + DAF system | 30,000 |
| PSA nitrogen generator + buffer tank | 12,000 |
| Pyrolysis reactor, condenser, catalyst | 40,000 |
| Controls, sensors, PLC | 15,000 |
| Noise control package | 20,000 |
| **Total module cost** | 270,000 |
| Vault civil works (pre-cast, new deployment) | 40,000 |
| Vault adaptation works (retrofit, indicative) | 15,000–35,000 |
| Installation (crane, connections) | 15,000 |
| **Total installed cost (new deployment)** | 325,000 |

In volume manufacturing (≥50 units/year), module cost is expected to reduce to €180–220k. Retrofit civil works vary by site and require individual quotation.

## 6.2 Operating Benefits — Reference Case (200–250 Households, Per Year)

| Item | Value (€/year) | Notes |
|---|---|---|
| Avoided hauling & tipping | 13,500 | 225 households × 2.5 p/hh × 0.5 kg/p/d × €150/tonne |
| Liquid fuel sales | 5,400 | 30 kg/batch × 2 batches/day × 300 days × €0.30/kg |

| | | |
|---|---|---|
| Recyclates (metals, glass) | 800 | |
| Organic broth to biogas | 1,200 | Value to receiving AD facility; reduces regional grid electricity |
| **Total annual benefit** | 20,900 | |

Payback period (scaled production, reference case): 8–12 years, strongest where hauling distances are long and landfill tipping fees are high.

**NOTE** *For a 50-household retrofit deployment, total annual benefit falls to approximately €5,000–6,000, extending payback well beyond 20 years. At this scale the economic case relies on grant support, avoided infrastructure investment, or premium tipping fee environments. The UWR is most economically viable at the 200+ household scale.*

# 7. Safety & Environmental Impact

## 7.1 Hazardous Item Mitigation

- Camera accountability at the lid deters and retrospectively identifies hazardous deposits.
- Inlet size restriction prevents insertion of large hazardous items.
- Metal detector with reject flap diverts batteries and gas cartridges to a locked box; CO sensor and motor current anomaly detection provide early warning of thermal events.
- Separate locked hazardous waste bin at the site for batteries, gas cartridges, and chemicals.
- Emergency shutdown interlocks and nitrogen flood on confirmed thermal event.

## 7.2 Pressure Vessel & Pyrolysis Safety

- Steam vessel certified to PED/ASME with redundant safety valves.
- Wash-filter box connected to expansion chamber via dedicated pressure relief path; overpressure from valve failure vents safely rather than into the open enclosure.
- Pyrolysis module in ATEX-rated enclosure.
- Nitrogen inerting throughout dryer, airlock, and reactor.

## 7.3 Environmental Controls

- Flue gas: wet scrubber removes HCl, particulates, and trace organics.
- Wastewater: closed-loop MBR + DAF system; sewer discharge limited to emergency overflow.
- Char: characterized for heavy metals before disposal; hazardous char handled by licensed contractor.

### 7.4 Noise & Vibration Compliance

Staged decompression, isolation mounting, acoustic enclosure, and operational scheduling combine to target compliance with residential night-time limits (35–45 dB(A) at facade).

## 8. Regulatory Pathway

- Pressure equipment: PED certification; notified body inspection.
- ATEX/IECEx: All electrical equipment in explosive atmosphere zones certified.
- Environmental permit: Required for pyrolysis operation.
- Waste management license: Required for MSW processing; obtainable under local regulations for small-scale decentralized facilities.
- Pyrolysis oil end-of-waste status: End-of-waste classification under the EU Waste Framework Directive (2008/98/EC) is a regulatory process to be completed during the pilot phase, not an existing approval. A quality specification meeting fuel oil standards will be developed and submitted during pilot operation as the basis for this application.
- Urban planning: Above-ground enclosure or support house requires planning permission; community consultation and environmental impact assessment may be required.
- Retrofit sites: May require additional planning consent and structural assessment of the existing vault.

## 9. Prototype Test Objectives

The following empirical questions are the primary objectives of the prototype phase, in priority order:

21. Throughput validation — confirm actual plastic yield per batch at the reference service population.
22. Water balance — close the water balance under operating conditions; confirm MBR and buffer tank sizing against measured condensate and wash water volumes.
23. Condensate & broth characterization — measure chlorine content, heavy metals, and COD in steam explosion condensate and organic broth; determine whether upstream pre-treatment is required before the MBR.
24. NIR performance on steam-exploded feedstock — measure false rejection rate and PVC carryover; determine whether standard NIR is sufficient or an upgrade is warranted.
25. Lip gap separator calibration — establish vacuum level and conveyor speed settings for clean plastic/glass separation; evaluate whether knife mill self-aspiration is sufficient or a dedicated vacuum source is needed.
26. Ultrasonic threshold calibration — establish the volume reduction ratio that reliably predicts adequate treatment across seasonal waste composition variation.
27. Energy balance validation — measure actual steam consumption, waste heat recovery fractions, and net external energy demand against the model.
28. Noise & vibration compliance — measure acoustic output at vault surface under full operational scheduling.

# 10. Implementation Roadmap

29. Finalize 3D CAD & docking design — confirm spatial feasibility for both new-deployment and representative retrofit envelopes; resolve geometry of pressure vessel / wash-filter box interface and inlet door chute-fold clearance.
30. Build prototype — construct one module; test at a controlled facility with real MSW against the objectives above.
31. Pilot installation — deploy at a single site (industrial zone or municipal yard) with full regulatory oversight.
32. Refine based on data — adjust MBR capacity, energy recovery, noise control, and contaminant management as indicated by prototype results.
33. Volume manufacturing — produce standardized modules; pre-cast vaults for new sites; develop a retrofit assessment toolkit for existing sites.
34. Deployment — roll out to neighborhoods with site-specific noise, vibration, and vault condition assessment.

# 11. Conclusion

Version 8 of the Underground Waste Refinery is the final concept design ready for prototype engineering. The document resolves all principal technical inconsistencies identified in earlier versions and incorporates five targeted clarifications: the wash-filter box now includes an explicit pressure relief path to the expansion chamber; the passive hinged chute fold mechanism is clearly specified; the organic broth baseline disposition is defined as transport to a centralized AD facility, with on-site AD as an optional add-on; the pyrolysis oil end-of-waste status is correctly framed as a pilot-phase regulatory objective; and the economic case for small retrofit deployments is explicitly acknowledged as dependent on local conditions and support mechanisms.

The separation train has been redesigned around a single continuous conveyor system that spreads, sorts, concentrates, and separates material through geometry and low-energy mechanisms — replacing a full pneumatic separation column with a V-convergence lip gap deflector at a fraction of the energy cost. The wash-filter box is a clearly defined atmospheric fabricated steel enclosure directly coupled to the pressure vessel discharge. The pressure vessel loading and discharge sequence is entirely gravity-assisted, minimizing actuators and moving parts. AI monitoring is split logically between ultrasonic volume reduction on the pressure vessel and camera-based adaptive feedback on the wash-filter box, giving the system genuine self-calibration capability.

All remaining open items are prototype test objectives, not design blockers. The UWR offers a viable, low-carbon, distributed alternative to conventional waste management — converting a municipal cost center into a local resource recovery asset.

## Acknowledgements

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